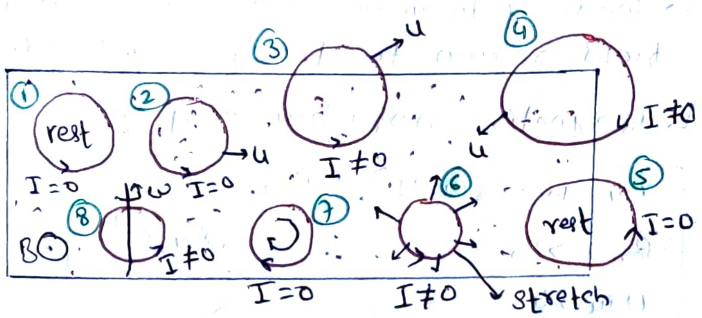
Electromagnetic Induction



Magnetic flux

↳ Counting of Magnetic field lines passing through given area.

- Scaler
- Dimension → $[ML^2T^{-2}A^{-1}]$
- S.I unit → Tesla-m^2 = weber (wb)
- C.G.S unit → Gauss-cm^2 = Maxwell
- 1 wb = 10^8 maxwell

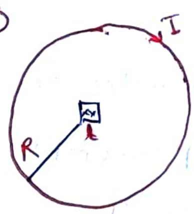
$$\boxed{\phi = \vec{B}\cdot\vec{A} = BA\cos\theta}$$

θ = Angle b/w Mag. field & Area.

Q → $\vec{B} = B_1\hat{i} + B_2\hat{j} - B_3\hat{k}$, $A = \vec{S}$

(i) x-y plane	(ii) y-z plane	(iii) z-x plane
$A = S\hat{k}$	$A = S\hat{i}$	$A = S\hat{j}$
$\phi = -B_3S$	$\phi = B_1S$	$\phi = B_2S$

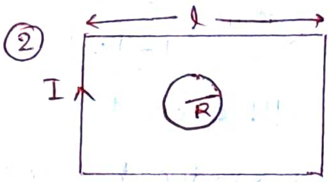
Q → ①



(R >> l)

$$\boxed{\phi = BA = \frac{\mu_0 I}{2R} l^2}$$

②



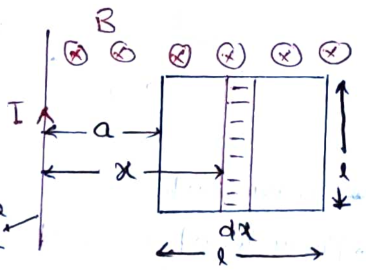
(R << l)

$\phi = B_{sq} \times \pi R^2$

$$\boxed{\phi = \frac{2\sqrt{2}\mu_0 I}{\pi l}\pi R^2}$$

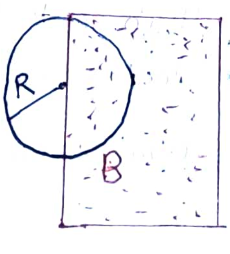
③ Flux through square loop?

$$\boxed{\phi = \frac{\mu_0 I l}{2\pi}\log\left(\frac{a+l}{a}\right)}$$



④ Flux through circular loop?

$$\boxed{\phi = B\left(\frac{\pi R^2}{2}\right)}$$



$$\boxed{\phi = \vec{B}\cdot\vec{A} = BA\cos\phi}$$

ϕ = const	ϕ = variable
Iin = 0	Iin ≠ 0

- * dirⁿ of Induced current
- * Lenz law
- * value of induced current
- * faraday's law

Lenz law

→ when flux is variable through any loop then direction of Induced current is such that it will oppose the cause due to which it is induced.

MR* law :-

↳ coil → flux का inertia में होता है। then coil will oppose the change, by induced current.

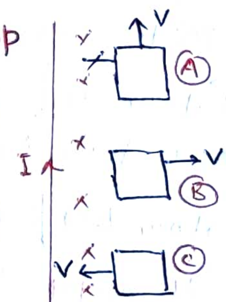
External field		Induced field
• B⊙ ↑	⟶	Bin⊗ (Iin = CW)
• B⊙ ↓	⟶	Bin⊙ (Iin = ACW)
• B⊗ ↑	⟶	B⊙ (Iin = ACW)
• B⊗ ↓	⟶	B⊗ (Iin = CW)

current induced in loop

A → zero (ϕ⊗ = const)

B → CW (B⊗↓ → B⊗)

C → ACW (B⊗↑ → B⊙)



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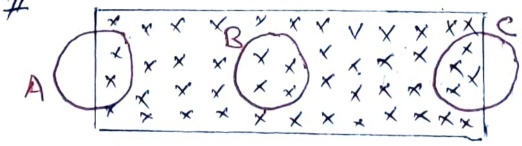

current induced in loop → Ⓐ → ACW Ⓑ → 0 Ⓒ → CW

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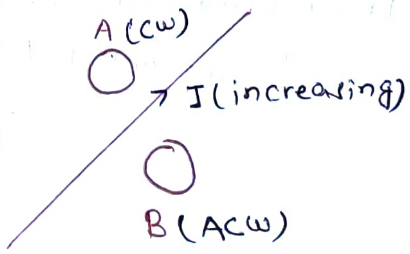
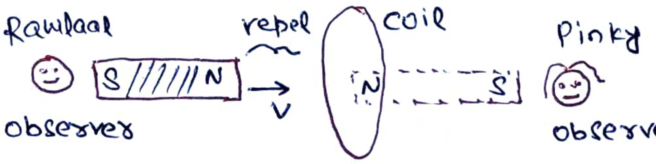


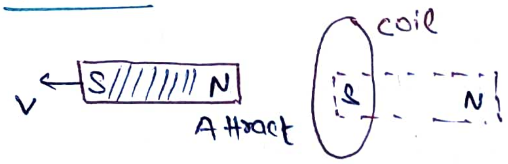
Case:-1



↳ current induced in loop
- w.r.t Ramlal → ACW
- w.r.t Pinky → CW

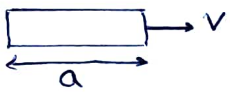
MR* → Aage se Anti then Picche se clock.

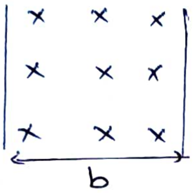
Case:- II



Q ⇒ Find time for which current will induced in rectangular loop?

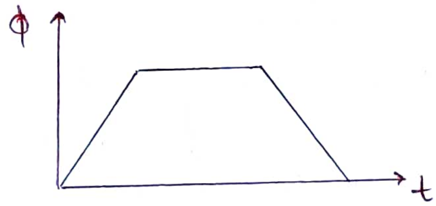
(i) a > b $\quad t = \frac{2b}{v}$

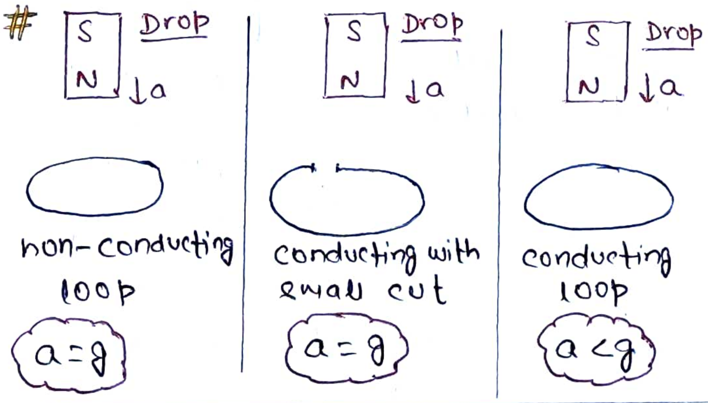




(ii) a < b $\quad t = \frac{2a}{v}$

variation of flux through the loop with respect to time:-





Faraday's law of EMI

↳ Based on conservation of energy

↳ gives value of induced e.m.f & current.

$\phi = BA\cos\theta$

At any instant

- $\epsilon.m.f_{inst} = -\frac{d\phi}{dt}$

– sign gives direcⁿ

- $|emf| = \frac{d\phi}{dt}$
- $I = \frac{emf}{R} = \frac{\Delta\phi}{R\Delta t}$
- $emf = \frac{d(BA\cos\theta)}{dt}$

In any time interval

- $|emf|_{Avg} = -\frac{\Delta\phi}{\Delta t}$
- $emf = \frac{\phi_f - \phi_i}{\Delta t}$
- $I_{Avg} = \frac{\Delta\phi}{R\Delta t}$
- $\Delta Q = \frac{\Delta\phi}{R}$

Motional Electromotive force

① Rail Problem

$\boxed{I = \frac{emf}{R} = \frac{Blv}{R}} \rightarrow |emf| = Blv$

$I = \frac{Blv}{R}$ $\quad \boxed{F = (Bl)I = \frac{B^2l^2v}{R}}$

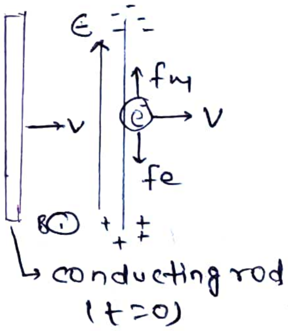
$\boxed{Power = \frac{B^2l^2v^2}{R}}$

② Hall effect

$f_M = f_e$ (After some time)

$evB = eE$

$\boxed{\Delta V = BVl_{eff}}$



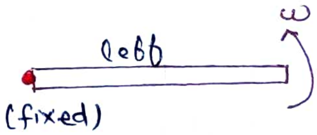
Scaler Triple product:-

$\epsilon = (\vec{V} \times \vec{B}) \cdot \vec{l}$

★ direcⁿ of EMF:-

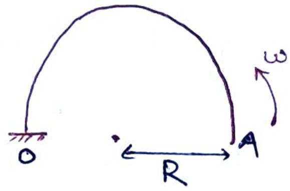
- Palm → velocity
- curl finger → Magnetic field
- Thumb → EMF

$\boxed{Emf = \frac{1}{2}B\omega(l_{eff})^2}$



$\boxed{Emf_{OA} = \frac{1}{2}B\omega(2R)^2}$

$l_{eff} = 2R$



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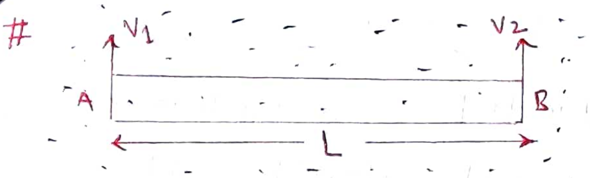


$\boxed{(Emf)_{AB} = B\left(\frac{\vec{V_1}+\vec{V_2}}{2}\right)L}$ → Pure transl^n motion

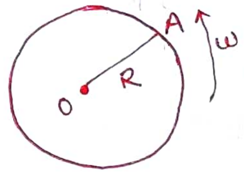
$\vec{V_1} = \vec{V_2} = V$

emf = BVL

Pure Rot^n Motion

$V_1 = 0$, $V_2 = L\omega$ → emf $= \frac{1}{2}BL^2\omega$

#

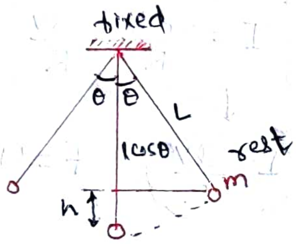


$\mathcal{E}_{OA} = \frac{B\omega R^2}{2}$

* Disc → collection of infinite rods.

#

Emf $= BL\sin\left(\frac{\theta}{2}\right)\sqrt{gL}$



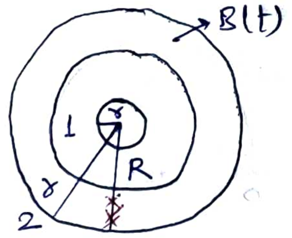
Induced Electric field

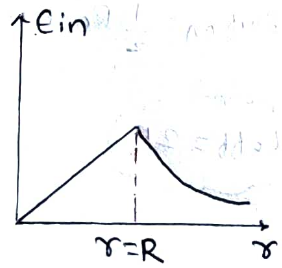
↳ Only time varying magnetic field induced electric field.

Electric field

असली	नकली
• Due to rest charge.	• Due to time varying magnetic field
• Electrostatic field.	• Induced electric field.
• Conservative field	• Non-conservative field.
• $\oint \vec{E}\cdot d\vec{l} = 0$	• $\oint \vec{E}\cdot d\vec{l} \neq 0$
• Does not form close loop.	• Always forms close loop.
	• only circular close loop for time varying magnetic field in cylindrical region

value of induced Electric field :-





$\boxed{\underset{(in)}{\mathcal{E}_1} = \frac{r\,dB}{2\,dt}}$ $\boxed{\underset{(out)}{\mathcal{E}_2} = \frac{R^2}{2r}\frac{dB}{dt}}$

Self induction and Solenoid

$\vec{I}$ (L, r, N, n) $\left(n = \frac{N}{l}\right)$

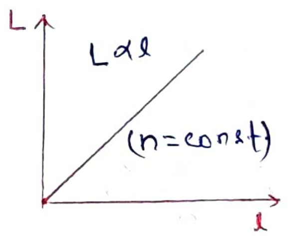
$\boxed{\phi = \mu_0 n^2 A l I}$ $\boxed{L = \mu_0 n^2 A l = \frac{\mu_0 N^2 A}{l}}$

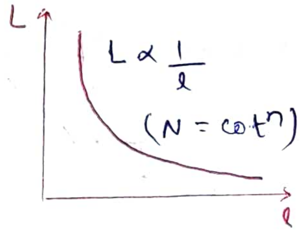
depends on solenoid.

$\boxed{\phi = LI}$ → self inductance depends only on solenoid not on current & flux.

$\boxed{L = \frac{\phi}{I} = (\mu_0 n^2 A l)}$ ↳ Scalar, $[ML^2T^{-2}A^{-2}]$

↳ Unit → $\frac{Wb}{Amp}$ = Henry





• If Iron placed inside solenoid → self inductance ↑ses. $L = \mu_{iron} n^2 A l$

⇒ At an instant, flux remains same.

$\phi = cost^n = LI$ $\boxed{L_1I_1 = L_2I_2}$

#

I = const B = const, ϕ = const] Emf = 0

#

I ↑ ϕ ↑ —|⊢— Emf $= L\frac{dI}{dt}$

#

I ↓ ϕ ↓ —||— Emf $= L\frac{dI}{dt}$

Magnetic Energy stored in Solenoid / Magnetic Energy density / Power loss

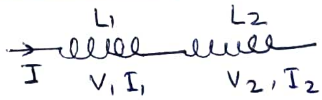
$\boxed{\underset{loss}{\text{Power (P)}} = VI = LI\frac{dI}{dt}}$ Analogy in electrostatics

$\boxed{\text{Magnetic energy } (\mathcal{E}) = \frac{1}{2}LI^2}$ → $\mathcal{E} = \frac{1}{2}CV^2$

$\boxed{\text{Magnetic energy density } (U) = \frac{B^2}{2\mu_0}}$ → $U = \frac{1}{2}\epsilon_0 E^2$

Series combination of Inductor

$\boxed{I = I_1 = I_2 \text{ (same)}}$



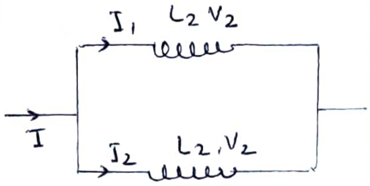
$\boxed{V = V_1 + V_2}$ $\boxed{L_{eq} = L_1 + L_2}$

Parallel combination of Inductor

$\boxed{V = V_1 = V_2 \text{ (same)}}$

$\boxed{I = I_1 + I_2}$

$\boxed{\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2}}$



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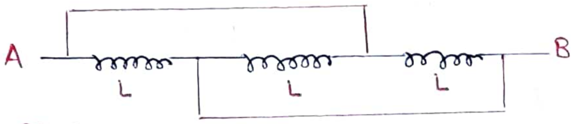
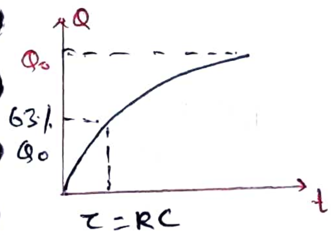


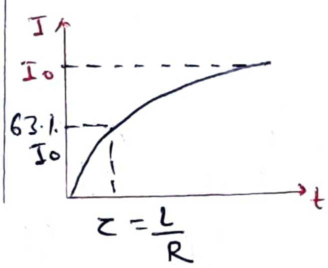
$L_{AB} = \frac{L}{3}$

Charging of Inductor

$t=0$	$t=\infty$ (steady state)
I = open wire C = simple wire	I = simple wire C = open wire
• $V_L = L\left(\frac{dI}{dt}\right) = \epsilon$ • $V_R = 0$	• $V_L = 0$ • $V_R = \epsilon$ • $I = \epsilon/R$ • $\epsilon_{Max} = \frac{1}{2}L\left(\frac{\epsilon}{R}\right)^2$

Capacitor	Inductor
$Q = Q_0[1-e^{-t/RC}]$	$I = I_0[1-e^{-Rt/L}]$
$\tau = RC$	$\tau = L/R$
• $t=0$ $Q=0$	• $t=0$ $I=0$
• $t=\infty$ $Q=Q_0$	• $t=\infty$ $I=I_0$





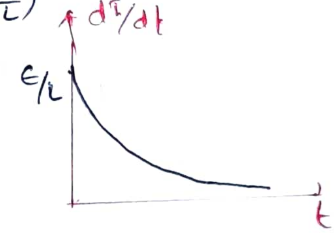
Graph b/w $\frac{dI}{dt}$ with Time

Rate of change in current:-

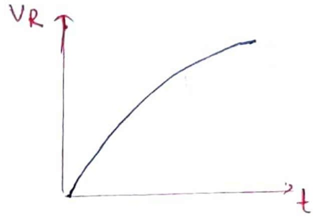
$$\frac{dI}{dt} = \left(\frac{\epsilon}{L}\right)e^{-t/\tau}$$

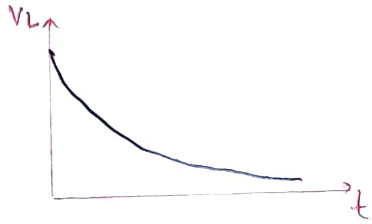
$t=0$ $\qquad$ $t=\infty$

$\frac{dI}{dt} = \left(\frac{\epsilon}{L}\right)$ $\qquad$ $\frac{dI}{dt} = 0$



$V_R = I_0(1-e^{-t/\tau})R$ $\qquad$ # $V_L = L\left(\frac{dI}{dt}\right) = \epsilon e^{-t/\tau}$



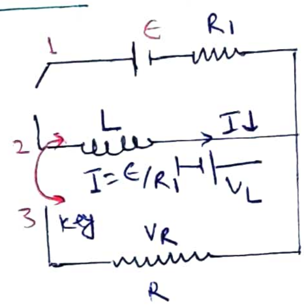


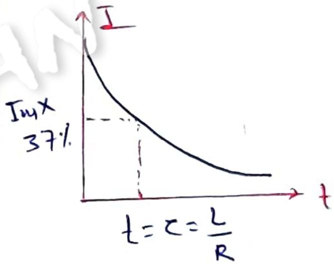
Discharging of Inductor

"$V_L = V_R$"

$\boxed{Q = Q_0 e^{-t/RC}}$

$\boxed{I = I_{max} e^{-Rt/L}}$

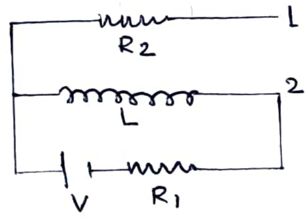




After complete charging of inductor key is shifted from 1 to 3.

⇒ Find total heat loss across R_2 when key is shifted from (1) to (2):-

$$\boxed{\begin{array}{l}\epsilon_{loss} \\ \text{or} \\ \epsilon_{stored}\end{array} = \frac{1}{2}\frac{LV^2}{R_1^2}}$$



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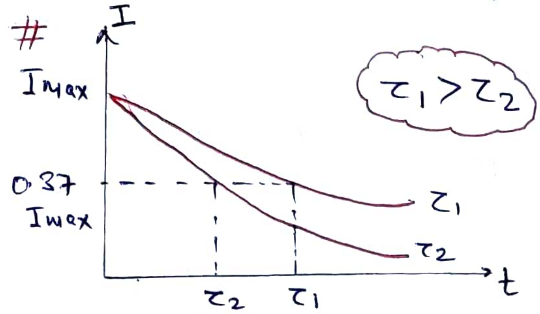
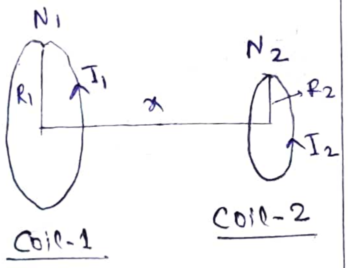


$\tau_1 > \tau_2$

Mutual Inductance

$\phi_1 = M I_2$, $\phi_2 = M I_1$

$$M = \frac{\mu_0 \pi R_1^2 R_2^2 N_1 N_2}{2x^3}$$



↳ Mutual inductance does not depend on flux and current.

Reciprocity theorem :-

$\boxed{M_{12} = M_{21}}$ → Always true

$$\boxed{\begin{aligned} \phi_{12} &= M_{12} I_2 , & \phi_{21} &= M_{21} I_1 \\ \varepsilon_{12} &= M_{12}\frac{dI_2}{dt} , & \varepsilon_{21} &= M_{21}\frac{dI_1}{dt} \end{aligned}}$$

★ Note:– Coupling factor

$M = K\sqrt{L_1 L_2}$ K = coupling factor

- K = 1 for perfect coupling
- K = 0 for No coupling
- 0 < K < 1 for imperfect coupling

Series combination of Inductor considering mutual induction :-

$$\boxed{L_{eq} = L_1 + L_2 \pm 2M}$$

- when flux link with same order = +ve
- when flux with opposite dir^c = –ve

Eddy current :-

↳ Agar किसी तार में current flow तो उसे kehte हैं current और plate में current flow करेगा, तो Eddy current.

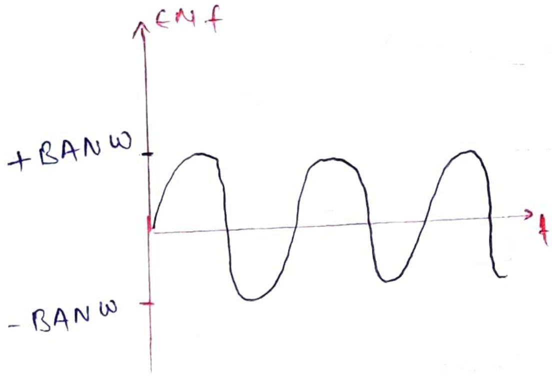
Application of Eddy currents

(i) Magnetic braking in trains
(ii) Electromagnetic damping
(iii) Induction furnance
(iv) Electric power meters.

AC Generator

$\phi = BAN\cos(\theta)$, $[\theta = \omega t]$

$$\text{Emf} = BAN\omega \sin(\omega t)$$



$|\text{Emf}|_{max} = BAN\omega \sin(\omega t)_{max}$

$$|\text{Emf}|_{max} = BAN\omega$$